Project Proposal

Land Deed Fraud & Dispute Risk Checker

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	15
Clarity	3					
Content Quality	6					
Spelling & Grammar	3					
Organization and Formatting	3					
Total obtained mark						
Comments						

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Table of Contents

1. Introduction

1.1 Background Overview

1.2 Problem Statement

1.3 Objectives

1.4 Scope

1.5 Stakeholders

1.6 Existing System

1.7 Proposed Solution

2. System Requirements

2.1 Functional Requirements

2.2 Non-Functional Requirements

2.3 Ethical, Legal, and Social Considerations

3. Tools and Technologies

3.1 Tools and Technologies used

3.2 System architecture

4. Project Timeline and Work Plan

4.1 Timeline and workplan

4.2 Future Work

References

1. Introduction

Land ownership in Bangladesh is governed by a layered and largely paper-based documentation system in which a single parcel of land is referenced across several independent records, including the sale deed, the khatian (record of rights), and the mutation or namjari certificate. Because these records are generated and stored by different offices and are rarely cross-checked against one another in a systematic way, the verification of land ownership before a purchase or inheritance transfer remains a slow, manual, and error-prone process. This proposal presents a software-based solution that applies optical character recognition, machine learning, and statute-grounded reasoning to automate this verification process and to help ordinary users identify irregularities before they become costly disputes.

The project is being undertaken as part of the Software Engineering Design Capstone Project course, which requires the development of a solution in which artificial intelligence performs work that a human currently carries out manually, rather than serving as a decorative addition to a conventional application. The land-document verification process described in this proposal satisfies this requirement directly: the core value of the proposed system depends entirely on its ability to read, extract, cross-reference, and explain document content in a way that a land office clerk or a junior lawyer presently does by hand. Removing the artificial intelligence components would leave no usable product, which confirms that the proposed system is appropriately scoped for this course.

1.1 Background Overview

The application domain of this project is legal technology, specifically the sub-domain of land administration and property-document verification within the broader Law and Policy sector. Verifying a plot of land currently requires a buyer, heir, or legal representative to physically collect and compare documents from the sub-registry office, the union land office, and the relevant khatian records. This process depends heavily on the manual judgement of a land office clerk or a hired lawyer, who must check for consistency in plot numbers, ownership chains, and mutation history by reading printed and handwritten paperwork side by side.

Current practice treats each document as an isolated artefact. A sale deed is registered at a sub-registry office, a khatian is maintained separately by the settlement and land-records authority, and a mutation certificate is issued by the relevant union land office once a transfer is reflected in the record of rights. Because these three offices do not routinely share a unified digital record for a given plot, an individual verifying a transaction must visit each office in turn, request copies, and manually compare the plot number, area, and

ownership details recorded in each document. Where a discrepancy exists, such as an outdated mutation record or a plot number that has been altered between documents, detecting it depends entirely on the diligence and experience of whoever is performing the comparison.

No widely accessible digital tool currently performs this cross-referencing automatically or explains detected irregularities in relation to the actual statutory provisions that govern land transactions in Bangladesh. The limited digitisation efforts that do exist within land administration are primarily focused on digitising individual record types for internal office use, rather than on providing an end-user-facing tool that reasons across multiple document types at once. This gap is the specific opening that the proposed project is designed to address.

1.2 Problem Statement

Before completing a land purchase or accepting an inherited property, an ordinary citizen or a junior legal practitioner must manually determine whether a deed is genuine, whether the plot has been sold to more than one buyer, whether the mutation record matches the deed, and whether any unresolved dispute is attached to the plot. This verification is currently performed by manually comparing multiple paper documents, a process that is time-consuming, dependent on individual expertise, and prone to oversight, particularly when fraud patterns such as duplicate sale or mismatched plot identifiers are deliberately concealed within dense or handwritten text.

Three specific limitations in the existing process motivate this project. First, there is no systematic cross-document consistency check: a buyer who obtains a deed and a khatian separately has no straightforward way to confirm that the two documents actually describe the same plot under the same ownership chain. Second, there is no accessible explanation layer: even when a knowledgeable reviewer notices an inconsistency, translating that observation into a specific, citable legal basis requires familiarity with statutes that most buyers do not have and most junior practitioners must look up manually. Third, the volume and format of these documents, many of which combine printed headers with handwritten entries, makes the manual review process slow enough that it is frequently skipped altogether under transactional time pressure, leaving buyers exposed to exactly the risks the review process is meant to catch.

Existing systems offer no structured way to cross-check these documents against one another or against the governing statutes, leaving buyers and junior practitioners to rely entirely on manual diligence. This gap disproportionately affects individuals who cannot afford extensive professional legal verification, and it contributes to land disputes that

frequently take years to resolve through the court system. The problem addressed in this proposal is therefore well scoped to a specific, verifiable task: automatically extracting the key fields from a limited set of common land document types, checking these fields for internal consistency, and explaining any detected inconsistency in plain language with reference to the specific statutory provision it relates to.

It is important to state clearly what this project does not attempt to solve. The proposed system is not intended to replace the official registration or mutation process, nor is it intended to issue a legal verdict on the validity of a transaction. It is designed strictly as a preliminary, evidence-based risk indicator that narrows the set of cases warranting closer professional or official scrutiny, which keeps the stated problem realistic and achievable within a single academic semester.

1.3 Objectives

General Objective: To design and develop a web-based decision-support system that automatically extracts information from scanned or photographed land documents, cross-references this information across related documents, and generates a plain-language risk report that flags potential fraud or ownership disputes with reference to the relevant provisions of Bangladeshi land law.

In support of this general objective, the project pursues the following specific objectives:

- To build a document intake and optical character recognition pipeline capable of processing printed and, to a limited extent, handwritten Bangla land documents.
- To fine-tune an existing pretrained model on a small set of collected sample documents in order to improve field-level recognition accuracy beyond a generic OCR baseline.
- To implement a structured data extraction layer that converts recognised text into defined fields such as plot number, khatian number, owner name, mouza, and transaction date.
- To implement a rule-based cross-document matching engine capable of detecting a defined set of common inconsistencies, including mismatched plot numbers, incomplete mutation chains, and duplicate-sale indicators.
- To implement a retrieval-augmented generation pipeline that grounds every flagged inconsistency in a specific, citable provision from the relevant statutes.
- To evaluate the resulting system with the support of a practising legal or land-office professional and to document its accuracy and limitations honestly.

Each specific objective above corresponds directly to one stage of the system pipeline described in Section 1.6, which allows project progress to be measured objectively against a working component at each stage of the semester rather than against a single, all-or-nothing final deliverable.

1.4 Scope

The in-scope features of the proposed system are listed below:

- Upload of scanned images or PDF files of deeds, khatian records, and mutation (namjari) papers.
- Optical character recognition of printed Bangla text, supported by a fine-tuned model for a limited set of common handwriting patterns.
- Extraction of a defined set of structured fields from each uploaded document.
- Cross-document consistency checking limited to an initial set of two to three well-documented fraud patterns: duplicate sale of the same plot, mismatched plot or khatian numbers across documents, and incomplete or missing mutation steps.
- Generation of a plain-language risk report, in Bangla, that cites the specific statutory provision relevant to each flag using RAG pipeline, accompanied by a clear disclaimer directing the user to verify findings with a qualified professional or the relevant land office.
- Role-based report views for general buyers and for legal or land-office professionals.
- A pilot scope limited to the document format and record-keeping conventions of a single district, selected in consultation with the project's interviewed domain contact.

The following features are explicitly out of scope for the current phase of the project:

- Direct, real-time integration with any official government land registry database or application programming interface.
- The issuance of a binding legal opinion, certification, or verdict of any kind.
- Support for the full range of document formats used across all districts of Bangladesh.
- Detection of fraud patterns beyond the initial defined set, including forged signatures or seals.
- A native mobile application; the system will be delivered as a web platform only.
- Processing of legal documents unrelated to land ownership, such as commercial contracts or tenancy agreements.

These boundaries are defined deliberately to prevent scope creep within a single academic semester and to ensure that every feature promised in this proposal is one the project team can realistically design, implement, and evaluate within the available time and resources. Any extension beyond these boundaries is addressed separately in Section 4.2, Future Work, so that it does not inflate the commitments of the current deliverable.

1.5 Stakeholders

The system is intended to serve and be governed by the following groups of stakeholders. Table 1 summarises each stakeholder's role and primary interest in the system, which is then described in further detail below.

Stakeholder	Role	Primary Interest
End Users	Prospective land buyers and heirs who upload documents before a transaction.	An affordable, understandable first check that reduces the risk of a fraudulent or disputed purchase.
Clients	Law firms, independent lawyers, and land-dispute consultants using the tool for client cases.	A faster, more consistent first pass over client documents before committing billable hours to manual review.
System Administrators	Project team members maintaining the models, data, and statutory knowledge base.	A maintainable, well-documented system that performs predictably and can be improved over time.
Regulatory / Organisational Bodies	Local sub-registry and union land offices consulted informally during research.	Confidence that any tool referencing land-administration procedures is accurate and does not overstate its authority.

Table 1: Stakeholder roles and interests.

End Users: Prospective land buyers and heirs who upload documents to assess the risk associated with a land transaction before committing to it. Their primary interest is an affordable, understandable first check that reduces the likelihood of entering into a fraudulent or disputed transaction. This group is expected to have little to no legal training, which directly shapes the requirement that all output be written in plain, non-technical Bangla.

Clients: Law firms, independent legal practitioners, and land-dispute consultants who may use the system as a preliminary screening tool for client cases. Their interest is a faster, more consistent first pass over client-supplied documents before committing billable hours to manual review, which is reflected in the more detailed, field-level report view designed specifically for this group.

System Administrators: The project team members responsible for maintaining the extraction models, the statutory knowledge base, and the overall reliability of the platform during and after the capstone period. Their interest is a maintainable, well-documented system that performs predictably and whose failure modes are easy to diagnose.

Regulatory and Organisational Bodies: Local sub-registry and union land offices, consulted informally during the research phase as a source of document samples and domain validation. These bodies hold no formal authority over the system and are not integrated with it directly; their interest lies in the accuracy of any tool that references land-administration procedures, which is why the system's output always directs the user back to official verification rather than substituting for it.

1.6 Existing System

At present, there is no single software system in use in Bangladesh that performs automated, cross-document verification of land ownership records in the manner proposed in this report. The closest existing practices fall into three categories, each of which addresses only part of the problem.

The first category consists of government digitisation initiatives that focus on digitising individual record types, such as khatian records, for internal administrative use. These efforts improve the storage and retrieval of individual documents within a single office but do not provide an end-user-facing tool that cross-references multiple document types for a single plot, nor do they interpret the legal significance of any detected inconsistency.

The second category consists of general-purpose legal information websites and applications, which provide access to the text of statutes or general legal guidance but do not process a user's specific documents or detect document-level inconsistencies. These tools are informational rather than diagnostic, and they place the entire burden of interpretation back on the user.

The third category consists of manual verification services offered by individual lawyers or land surveyors, which is the most functionally similar existing alternative to the

proposed system. This category correctly performs cross-document comparison and legal interpretation, but it does so manually, which is slow, inconsistent between practitioners, and often financially out of reach for buyers making smaller transactions. The proposed system is positioned to automate the core, repeatable part of this manual process, specifically the extraction and cross-referencing of structured fields, while explicitly preserving the role of a human professional for final verification and judgement.

1.7 Proposed Solution

The proposed solution is a web-based pipeline, organised as a coordinated sequence of automated stages, that takes a set of uploaded land documents and produces a plain-language, statute-grounded risk report. A user begins by uploading images or PDF copies of the deed, khatian, and mutation papers associated with a single plot of land.

The system first applies an optical character recognition engine to convert each document into machine-readable text. A generic OCR engine serves as the baseline for printed text, while a model fine-tuned on a small sample dataset collected during the research phase is used to improve recognition of document-specific vocabulary, abbreviations, and common handwriting patterns found in Bangla land records. Each recognised field is accompanied by a confidence score; where this score falls below a defined threshold, the system marks the field as requiring manual verification rather than presenting an uncertain reading as fact.

A structured data extraction layer then identifies and isolates the key fields needed for verification, such as the plot number, khatian number, owner name, mouza, and transaction date, converting unstructured recognised text into a consistent structured record for each document. Once each document has been reduced to a structured record, a cross-document matching engine compares the corresponding fields across all uploaded documents for the same plot and identifies inconsistencies belonging to the fraud patterns defined in the project scope, including duplicate-sale indicators, mismatched plot identifiers, and incomplete mutation chains.

Each detected inconsistency is passed to a retrieval-augmented generation pipeline, which searches an embedded knowledge base of the relevant statutory text, including the State Acquisition and Tenancy Act 1950 and the Registration Act 1908, and generates a plain-Bangla explanation that cites the specific clause responsible for the flag, rather than producing an unsupported or generic warning. The system finally compiles these flags into a single risk report, tiered by severity, and presents it through a role-appropriate interface:

a simplified summary for general buyers and a more detailed, field-level view for legal or land-office professionals. Every report carries a clear disclaimer directing the user to a qualified professional or the relevant land office for final verification. A high-level overview of this workflow is presented in Figure 1.

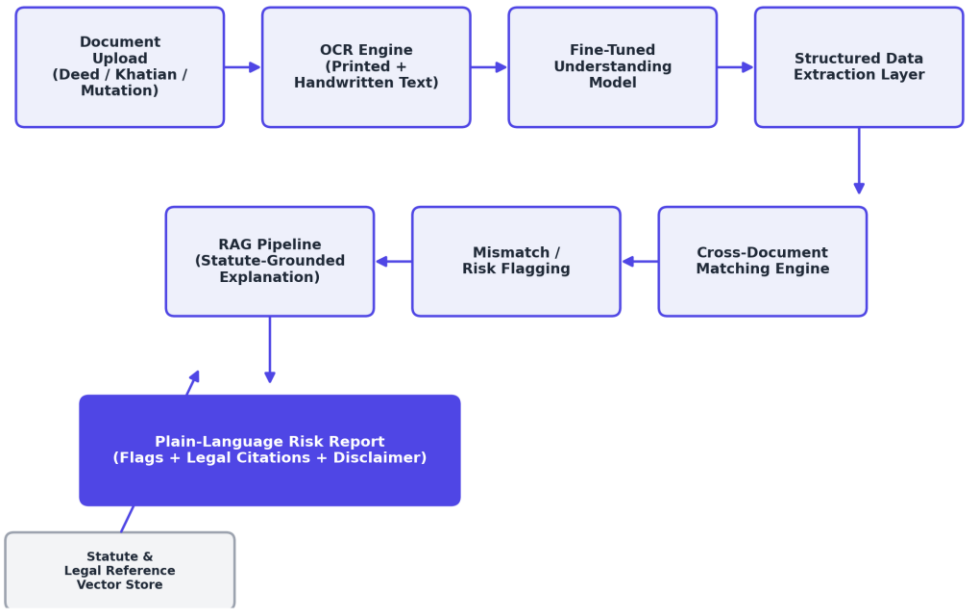


Figure 1: High-level system workflow of the proposed Land Deed Fraud and Dispute Risk Checker.

2. System Requirements

This section specifies the primary requirements that the proposed system is expected to satisfy. Requirements are grouped into functional requirements, which describe specific behaviours the system must perform, and non-functional requirements, which describe the quality attributes the system must exhibit. Each requirement is expressed as a clear, testable statement that can be independently verified during development and evaluation.

2.1 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow a registered user to upload one or more scanned images or PDF files of land documents for a single plot.	High

ID	Requirement	Priority
FR-02	The system shall convert uploaded document images into machine-readable text using an optical character recognition engine.	High
FR-03	The system shall extract a defined set of structured fields, including plot number, khatian number, owner name, mouza, and transaction date, from the recognised text of each document.	High
FR-04	The system shall compare extracted fields across all documents uploaded for the same plot and identify mismatches in plot number, khatian number, or ownership chain.	High
FR-05	The system shall detect duplicate-sale indicators when the same plot number appears in more than one unrelated deed submitted to the system.	High
FR-06	The system shall generate, for every detected inconsistency, a plain-language explanation that cites the specific statutory provision relevant to that inconsistency.	High
FR-07	The system shall assign a risk tier of low, medium, or high to each generated report based on the number and severity of detected flags.	Medium
FR-08	The system shall display a clear disclaimer on every generated report advising the user to seek verification from a qualified legal professional or the relevant land office.	High
FR-09	The system shall provide two distinct report views: a simplified summary view for general users and a detailed view, including raw extracted fields and confidence scores, for legal or land-office professionals.	Medium
FR-10	The system shall maintain a record of each uploaded case and its corresponding generated report for later retrieval by the user who submitted it.	Medium
FR-11	The system shall flag any document for which optical character recognition confidence falls below a defined threshold as requiring manual human verification, rather than silently presenting a low-confidence extraction as fact.	High

ID	Requirement	Priority
FR-12	The system shall allow an administrator to update or extend the embedded statutory reference text used by the retrieval-augmented generation pipeline.	Low

Table 2: Functional requirements.

2.2 Non-Functional Requirements

In addition to the functional behaviours listed above, the system is expected to satisfy the following quality attributes:

Performance: The system shall return a complete risk report for a standard three-document case (deed, khatian, mutation) within a reasonable processing time, targeted at under two minutes under normal load during the pilot phase.

Reliability: The system shall handle a failed or low-confidence OCR pass gracefully by flagging the affected document for manual review rather than terminating the process or silently producing an incorrect report.

Usability: The buyer-facing report view shall be understandable to a person without legal training, using plain Bangla and avoiding unexplained legal or technical terminology.

Security and Privacy: Uploaded documents and extracted personal data shall be stored with restricted access limited to the uploading user and authorised project administrators, and shall not be shared with any third party without explicit consent.

Maintainability: The statutory knowledge base used by the retrieval-augmented generation pipeline shall be stored separately from the application logic so that it can be updated or extended without requiring changes to the core codebase.

Auditability: Every generated flag shall be traceable to the specific extracted fields and the specific statutory clause that produced it, so that outputs can be reviewed and verified by a human evaluator.

Portability: The system shall be deployable through a containerised environment so that the development, testing, and demonstration environments remain consistent.

2.3 Ethical, Legal, and Social Considerations

Because the system processes documents containing personal and property-related information, uploaded files and extracted data will be handled with restricted access and will not be shared beyond the project team and, where applicable, the cooperating domain contact who consents to review anonymised outputs. Documents used for fine-tuning and testing will be anonymised or redacted wherever they are not already synthetic, and any identifying information not strictly necessary for the stated purpose of the project will be removed before storage.

The system is explicitly designed as a decision-support tool rather than a source of binding legal advice. Every generated report will carry a clear, prominent disclaimer stating that findings must be verified through a qualified legal professional or the relevant land office before any action is taken, and the project team will avoid any wording in the system's output that could be interpreted as a legal certification or guarantee.

Given the known limitations of handwriting recognition and the absence of a large labelled fraud dataset, the project team is committed to reporting the system's accuracy and failure cases honestly in the final project documentation, including clearly marking low-confidence extractions rather than presenting them as certain. The team also recognises a social responsibility to ensure that the system does not become a tool that could itself be misused, for example by someone probing it to learn which forgery patterns currently go undetected; access to the system during the pilot phase will therefore be limited to the project team, the course evaluators, and the cooperating domain contact.

3. Tools and Technologies

The following technologies have been selected for the design and implementation of the proposed system, based on their suitability for the respective components of the pipeline described in Section 1.6. Each category below also notes the reasoning behind the selection.

Programming Languages

- Python — used for the OCR pipeline, model fine-tuning, data extraction logic, and the retrieval-augmented generation pipeline, chosen for its mature ecosystem of machine learning and natural language processing libraries.
- JavaScript / TypeScript — used for the web-based client interface, chosen for its wide tooling support and the team's familiarity with modern front-end frameworks.

Frameworks and Libraries

- FastAPI — used to implement the backend application programming interface connecting the client to the processing pipeline, chosen for its asynchronous request handling, which suits the multi-stage, potentially slow document-processing workflow.
- React — used to implement the client-facing web interface, including the upload workflow and the role-based report views, chosen for its component-based structure, which maps naturally onto the two distinct report views required by FR-09.
- Tesseract OCR (with Bangla language support) and a cloud vision API — used as the baseline optical character recognition layer, selected to provide a working OCR capability from the start of development while the fine-tuned model is still being trained.
- Hugging Face Transformers and PyTorch — used to fine-tune a pretrained model on the project's collected sample document dataset, selected for their widespread academic use and compatibility with free-tier training environments such as Google Colab.
- LangChain or an equivalent orchestration library — used to implement the retrieval-augmented generation pipeline and the overall agent-style orchestration of the processing stages, selected to reduce the amount of custom orchestration code the team needs to write and maintain.
- FAISS or ChromaDB — used as the vector store for the embedded statutory reference text, selected for being lightweight, free to run locally, and sufficient for the scale of statutory text relevant to a single-district pilot.

Databases

- PostgreSQL — used as the primary relational database for user accounts, uploaded case records, extracted structured fields, and generated reports, selected for its reliability and strong support for structured, relational case data.
- A vector database (FAISS or ChromaDB, as noted above) — used specifically for similarity search over the embedded statutory text that grounds the retrieval-augmented generation pipeline.

Supporting Tools

- Git and GitHub — used for version control and collaborative development, with a complete commit history maintained as evidence of individual contribution, in line with the course's continuous evaluation requirement.
- Docker — used to containerise the application for consistent deployment during development, testing, and demonstration.
- A free-tier cloud hosting platform, such as Render or Vercel — used to host the development and demonstration deployment of the system at a cost appropriate for a student project.

3.2 System Architecture

The system follows a layered architecture consisting of a client layer, an application and orchestration layer, an artificial-intelligence processing layer, and a data layer, illustrated in Figure 2.

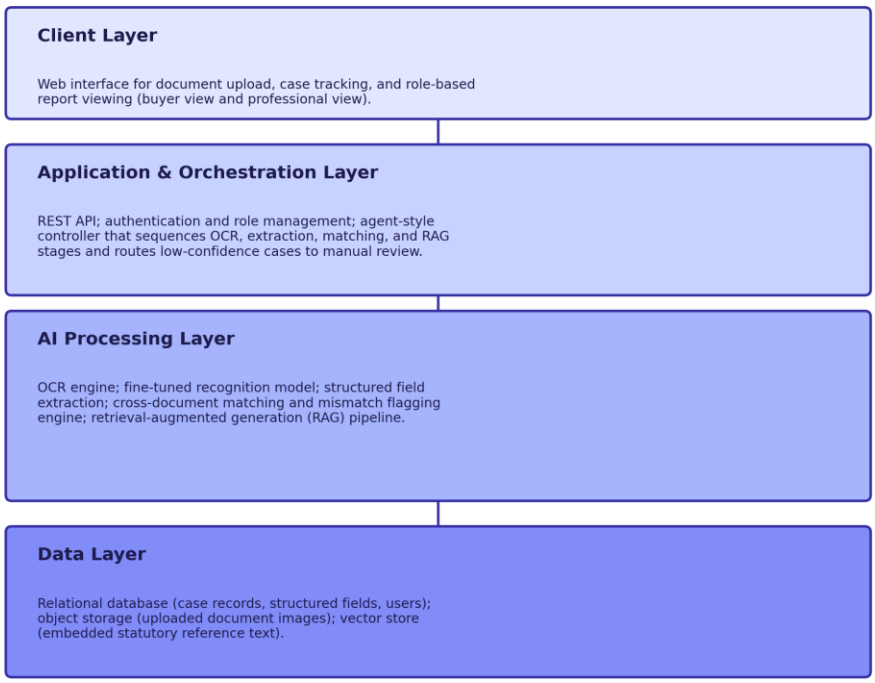


Figure 2: Layered system architecture.

The client layer is a responsive web interface through which users upload documents, track the status of a submitted case, and view generated reports in either the simplified buyer view or the detailed professional view. The application and orchestration layer exposes a

REST application programming interface and coordinates the sequence of processing stages, functioning as an agent-style controller that decides, at each step, whether to proceed to the next stage, retry a failed step with adjusted preprocessing, or route a document to manual review when optical character recognition confidence is too low to proceed automatically.

The artificial-intelligence processing layer contains the optical character recognition engine, the fine-tuned understanding model, the structured extraction logic, the cross-document matching and flagging engine, and the retrieval-augmented generation pipeline. This layer represents the core intellectual contribution of the project and is where the majority of development and evaluation effort will be concentrated. The data layer consists of a relational database for case records and structured fields, object storage for uploaded document images, and a vector store for the embedded statutory reference text. This layered separation allows each component to be developed, tested, and improved independently during the semester, and it allows the team to substitute or upgrade an individual component, such as the OCR engine, without requiring changes to the layers above or below it.

4. Project Timeline and Work Plan

The project is planned across the full fourteen-week semester timeline, structured around the milestone roadmap provided for this course. Table 3 presents the phase-wise breakdown of key deliverables and the team member primarily responsible for each phase, while recognising that testing, review, and documentation activities are shared across the full team throughout the project. Figure 3 presents the same plan as a week-by-week timeline chart, illustrating the deliberate overlap between the model fine-tuning phase and the matching engine phase, which allows the team to continue backend development while fine-tuning is in progress.

Phase / Weeks	Key Deliverables	Primary Responsibility
Weeks 1–2 Research & Scoping	Domain interview completed; sample documents collected; relevant statutes identified; final scope and fraud-pattern list confirmed.	All members

Phase / Weeks	Key Deliverables	Primary Responsibility
Weeks 3–5 Data Pipeline & OCR	Document upload pipeline; baseline OCR integration; preliminary structured extraction working end to end.	Member 2
Weeks 5–7 Model Fine-Tuning	Sample dataset labelled; existing model fine-tuned; extraction accuracy benchmarked against the generic baseline.	Member 1
Weeks 7–9 Matching Engine & Flagging	Cross-document matching logic implemented; mismatch flagging module completed and unit tested.	Member 2
Weeks 8–10 RAG Pipeline	Statutory text embedded into the vector store; retrieval-augmented explanation generation implemented and grounded.	Member 1
Weeks 9–11 Report Generation & UI	Role-based report views implemented; disclaimer and risk-tier logic finalised; client interface completed.	Member 3
Weeks 11–13 Testing & Validation	System tested against constructed and, where available, real anonymised cases; domain contact feedback incorporated.	Member 3
Weeks 13–14 Documentation & Demo	Final report, architecture diagrams, and demonstration prepared for submission and presentation.	All members

Table 3: Milestone-based project timeline and task distribution.

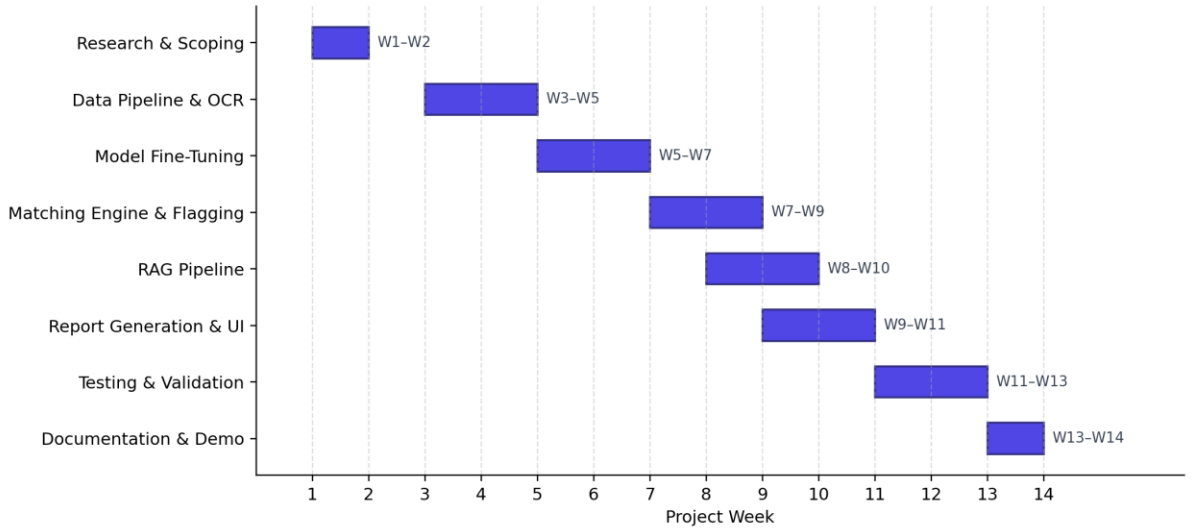


Figure 3: Week-by-week project timeline.

Task distribution is designed so that each member maintains an independently verifiable line of work, reflected in individual commit history, in keeping with the course's requirement for continuous, trackable contribution. Member 1 is primarily responsible for the data pipeline, OCR integration, and the matching and flagging engine. Member 2 is primarily responsible for dataset preparation and model fine-tuning. Member 3 is primarily responsible for the retrieval-augmented generation pipeline and the client-facing report interface. All members jointly participate in the initial scoping interview, in testing and validation, and in final documentation, and all members are expected to maintain weekly commits throughout the semester regardless of which phase is currently the primary focus of their individual work.

This distribution is intended as a starting allocation rather than a rigid division of labour. Because several phases overlap in time, team members are expected to assist one another where a task runs ahead of or behind schedule, and the weekly issue board maintained for the project will be used to track and reassign individual tasks as needed throughout the semester.

4.2 Future Work

Beyond the scope of the current capstone, the system could be extended to cover additional district record formats, to incorporate a larger and more diverse labelled dataset for the fine-tuned recognition model, and to explore a formal cooperation agreement with a sub-registry or union land office for a supervised pilot deployment beyond the academic semester.

A longer-term direction could involve extending the matching engine to detect a wider range of fraud patterns, such as inconsistencies in signatures or official seals, and integrating with official land registry systems, subject to the necessary institutional permissions, to move from a user-uploaded-document model toward direct record verification. A further direction worth exploring is a structured feedback loop in which outcomes from cases that proceed to formal dispute or verification are fed back into the system to progressively improve the accuracy of both the extraction model and the matching rules over time.

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